Yohan Cha

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Research interests: Scientific Machine Learning; physics-informed and data-driven modeling; uncertainty quantification; CFD: turbulence / multiphase / reacting flows; dynamical systems.

Education

Inha University — Incheon, South Korea
M.S., Mechanical Engineering (Thermal & Fluid Engineering) — Aug 2022
Advisor: Prof. Hyunchul Ju, Eco-Smart Power Lab

Inha University — Incheon, South Korea
B.S., Mechanical Engineering — Aug 2020

Publications

First Author

1. Choi, J.†, **Cha, Y.**†, Kong, J., Vaz, N., Lee, J., Ma, S.-B., Kim, J.-H., Lee, S. W., Jang, S. S., & Ju, H. (2022). Multi-Variate Optimization of Polymer Electrolyte Membrane Fuel Cells in Consideration of Effects of GDL Compression and Intrusion. *Journal of The Electrochemical Society*, 169(1), 014511.
2. **Cha, Y.**, Choi, J., & Ju, H. (2021). Pressure Swing Adsorption-Based Hydrogen Purification Vessel 3D Modeling and Feasibility Study. *Transactions of the Korean Hydrogen and New Energy Society*, 32(4), 197–204.

† Equal contribution.

Co-Author

3. Vaz, N., Choi, J., **Cha, Y.**, Kong, J., Park, Y., & Ju, H. (2023). Multi-objective optimization of the cathode catalyst layer micro-composition of polymer electrolyte membrane fuel cells using a multi-scale, two-phase fuel cell model and data-driven surrogates. *Journal of Energy Chemistry*, 81, 28–41.
4. Choi, J., Kim, E., **Cha, Y.**, Ghasemi, M., & Ju, H. (2022). Probing the influence of nonuniform Pt particle size distribution using a full three-dimensional, multiscale, multiphase polymer electrolyte membrane fuel cell model. *Electrochimica Acta*, 405, 139811.

Research & Professional Experience

SK Innovation — Daejeon, South Korea
CFD Research Engineer — Jan 2023 – Sep 2025
Process optimization across refining, petrochemicals, and battery manufacturing

- **Single Wall Carbon Nano Tube (SWCNT) Reactor Scale-Up**
  - Stabilized the lab-scale reactor process for SWCNT battery-material synthesis, extending equipment operating time by 6×, and scaled it up to commercial scale
  - Analyzed internal flow and temperature distribution as functions of gas flow rate, reactor geometry, and furnace heating conditions under high temperatures (>1300 °C)

- Implemented NIST-based thermal properties of H_2/N_2, species transport, and Discrete Ordinates (DO) radiation models in ANSYS Fluent

- **Polymer Autoclave Reactor Scale-Up**
  - Derived a scale-up design (reactor geometry, impeller geometry) for a high-pressure (>1300 bar) autoclave reactor to increase ethylene-acrylic acid (EAA) production
  - Established scale-up criteria by analyzing free-radical copolymerization, impeller-induced fluid behavior, and reaction heat inside the reactor
  - Modeled polymerization reactions (mass and heat conservation) using custom User-Defined Functions (UDFs) and impeller-induced mixing via the Multiple Reference Frame (MRF) approach in ANSYS Fluent
- **Three-Phase Transition Analysis in Refinery Process Piping**
  - Evaluated the feasibility of installing a new water-injection quill in Uni-cracking (UC) process piping, contributing to a 2× increase in process profitability
  - Analyzed three-phase transitions among liquid water, oil, and gas under high pressure (>150 bar) as water injection rates increased with process throughput, using a Raoult's-law-based formulation
  - Implemented Eulerian multiphase modeling in ANSYS Fluent with Raoult's-law-based phase-transition formulation

Doosan Mobility Innovation Yongin, South Korea
Fuel Cell Research Engineer Apr 2022 – Dec 2022

- **Air-Cooled Fuel Cell Flow-Field Design for Commercial Drones**
  - Designed anode/cathode flow fields and cooling channels for air-cooled PEMFCs targeting commercial drone applications
  - Improved water-removal efficiency over the reference flow field, enhancing overall energy efficiency, and achieved **10% improvement in cooling efficiency** by eliminating cooling dead zones
  - Developed a 3D PEMFC simulation model in ANSYS Fluent capturing H_2/O_2 flow, electrochemical reactions, and water/vapor two-phase transport

Inha University Incheon, South Korea
Graduate Research Assistant, Eco-Smart Power Lab (Advisor: Prof. Hyunchul Ju) 2020 – 2022

- **AI-Based Fuel Cell Flow-Field Optimization Platform**
  - Developed an end-to-end optimal design platform integrating data-driven surrogate models (Kriging, RSA, Multi-Layer Perceptron) with CFD analysis
  - Implemented the Python pipeline: DOE (Latin Hypercube Sampling) → PEMFC CFD simulation → AI-based surrogate training → optimal point selection via Genetic Algorithm
  - Findings published in *Journal of The Electrochemical Society* (see First Author, Publication 1)

Technical Skills

Programming Languages: C, Python, MATLAB.

Deep Learning Frameworks: PyTorch, JAX.

Numerical Solvers: ANSYS Fluent, OpenFOAM.

HPC & Systems: Linux-based HPC clusters — parallel ANSYS Fluent simulations, software installation/configuration, and system maintenance.

CAD Software: CATIA V5.

Microsoft Office: Word, PowerPoint, Excel.

Scholarships and Awards

- **Best Paper Presentation Award**, Korean Society for Fluid Machinery (KSFM) Jul 2021
  - Awarded for poster on optimal design of PEMFC flow fields considering GDL compression and intrusion
- **Advisor's Recommendation Scholarship**, Inha University Fall 2020 – Spring 2022

Teaching Experience

Inha University

- **Laboratory Teaching Assistant** – Physics Laboratory II Fall 2021
  - Taught physics fundamentals, experimental techniques, and data measurement and analysis (general education course)
- **Teaching Assistant** – Solid Mechanics Spring 2022
  - Q&A support and exam preparation for undergraduate students in Mechanical Engineering

Standardized Test Scores

TOEFL iBT (Jan 2026): 100

Languages

Korean (Native)
English (Professional Working Proficiency)

Military Service

Republic of Korea Army May 2016 – Feb 2018